

ICE4405P Digital Signal Processing

Lab 2 Report

DFT, FFT, and Signal Reconstruction

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Contents

1	Problem 1: Compute the DFT of an Audio Signal	3
1.1	Given	3
1.2	Result	3
1.3	Figure	4
2	Problem 2: Frequency Domain Modifications and Signal Distortion	4
2.1	Task	4
2.2	Method	4
2.3	Result	5
2.4	Discussion	5
2.5	Figure	5
3	Problem 3: Compute FFT and Compare Performance	5
3.1	Setup	5
3.2	Timing Result	6
3.3	Discussion	6
3.4	Figure	6
4	Problem 4: Effect of Zero-Padding on Spectral Resolution	7
4.1	Task	7
4.2	Observation	7
4.3	Figure	7
5	Problem 5: Low-Pass Filtering for Speech Signal Reconstruction	8
5.1	Given	8
5.2	Method	8
5.3	Discussion	8
5.4	Figures	9
6	Conclusion	10

1 Problem 1: Compute the DFT of an Audio Signal

1.1 Given

$$x[n] = [1, 2, 3, 4, 3, 2, 1, 0], \quad N = 8$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi kn}{N}}$$

1.2 Result

Computed DFT coefficients:

$$\begin{aligned} X[0] &= 16, \\ X[1] &= -4.8284 - 4.8284j, \quad X[2] = 0, \\ X[3] &= 0.8284 - 0.8284j, \quad X[4] = 0, \\ X[5] &= 0.8284 + 0.8284j, \quad X[6] = 0, \\ X[7] &= -4.8284 + 4.8284j. \end{aligned}$$

Magnitude spectrum:

$$|X[k]| = [16, 6.8284, 0, 1.1716, 0, 1.1716, 0, 6.8284]$$

Phase spectrum (rad):

$$\angle X[k] = [0, -2.3562, 0, -0.7854, 0, 0.7854, 0, 2.3562]$$

1.3 Figure

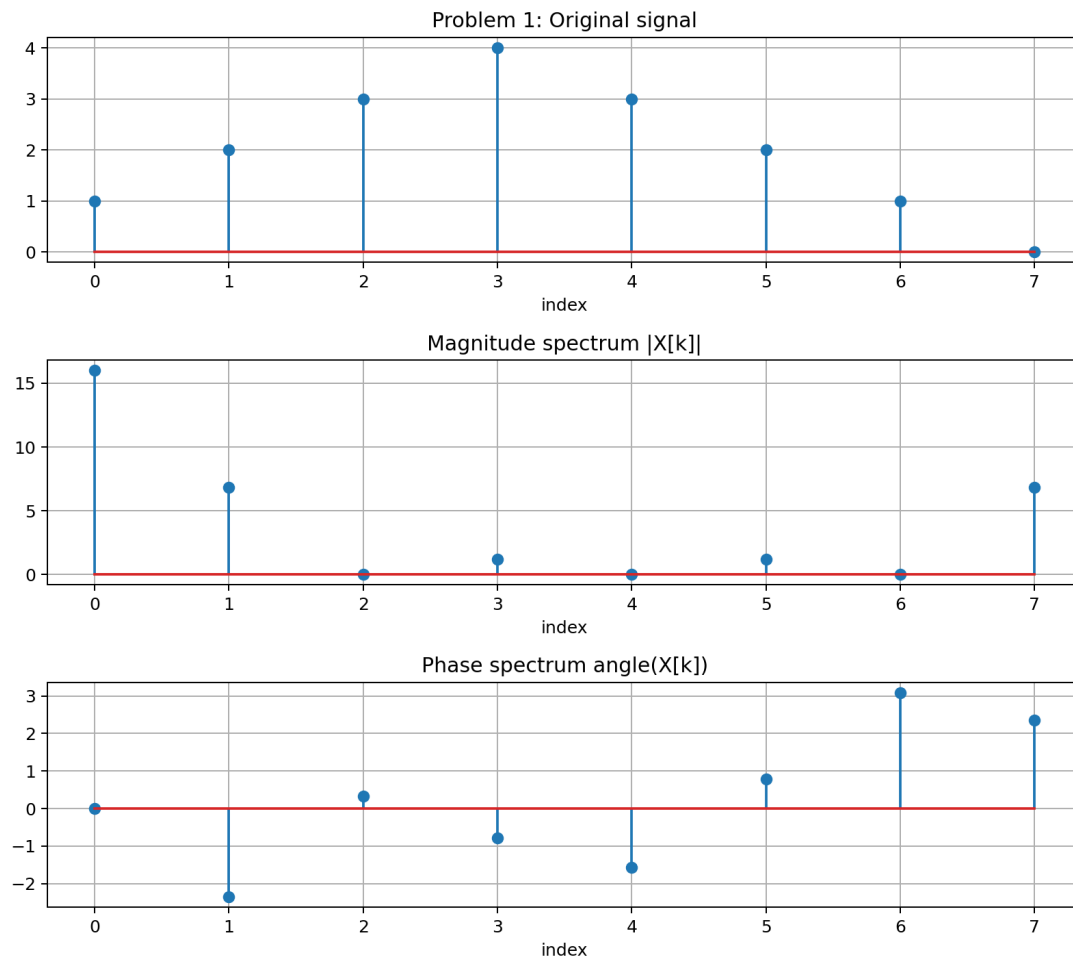


Figure 1: Original signal, magnitude spectrum, and phase spectrum

2 Problem 2: Frequency Domain Modifications and Signal Distortion

2.1 Task

Set all even-indexed DFT coefficients (in $k = 0, 1, \dots, N - 1$, i.e. even k) to zero, then compute IDFT.

2.2 Method

$$X_{\text{mod}}[k] = \begin{cases} 0, & k \in \{0, 2, 4, 6\} \\ X[k], & k \in \{1, 3, 5, 7\} \end{cases}$$

$$x_{\text{mod}}[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_{\text{mod}}[k] e^{j \frac{2\pi kn}{N}}$$

2.3 Result

$$x_{\text{mod}}[n] \approx [-1, 0, 1, 2, 1, 0, -1, -2]$$

(tiny numerical imaginary residue ignored)

2.4 Discussion

Keeping only odd-frequency bins introduces strong waveform distortion and changes amplitude pattern significantly. This is expected since half of the spectral components are removed.

2.5 Figure

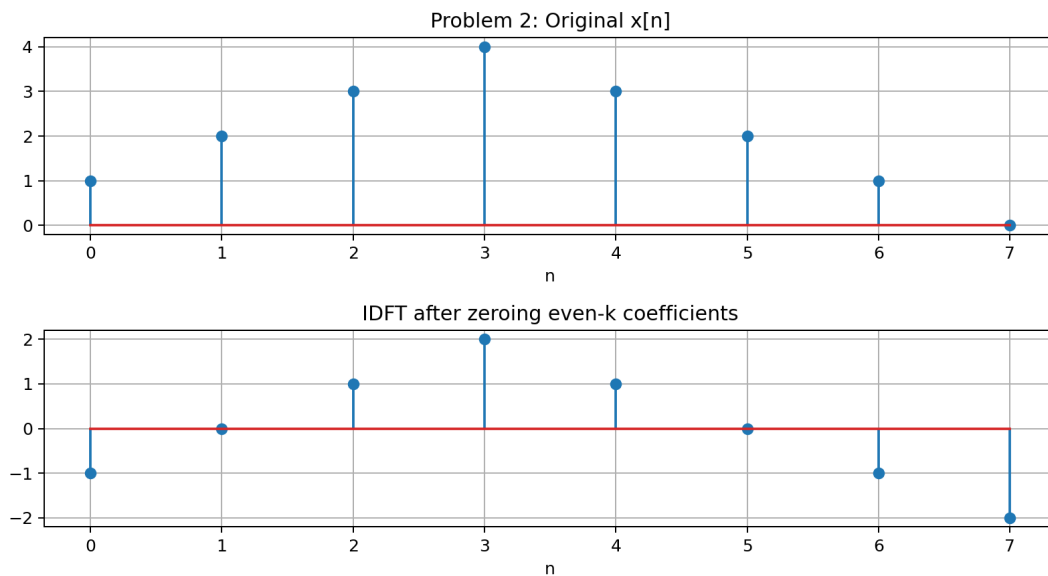


Figure 2: Original vs reconstructed signal after deleting even- k DFT bins

3 Problem 3: Compute FFT and Compare Performance

3.1 Setup

Use the same $x[n]$, zero-pad to $N = 1024$, and compare:

- Direct DFT (nested loops)
- FFT

3.2 Timing Result

$$t_{\text{DFT}} = 0.016490 \text{ s}, \quad t_{\text{FFT}} = 0.000024 \text{ s}$$

$$\text{Speedup} = \frac{t_{\text{DFT}}}{t_{\text{FFT}}} \approx 673.06 \times$$

3.3 Discussion

FFT provides a dramatic computational advantage for large N , while producing the same spectral result (up to numerical precision). This validates the $O(N \log N)$ efficiency gain over direct DFT.

3.4 Figure

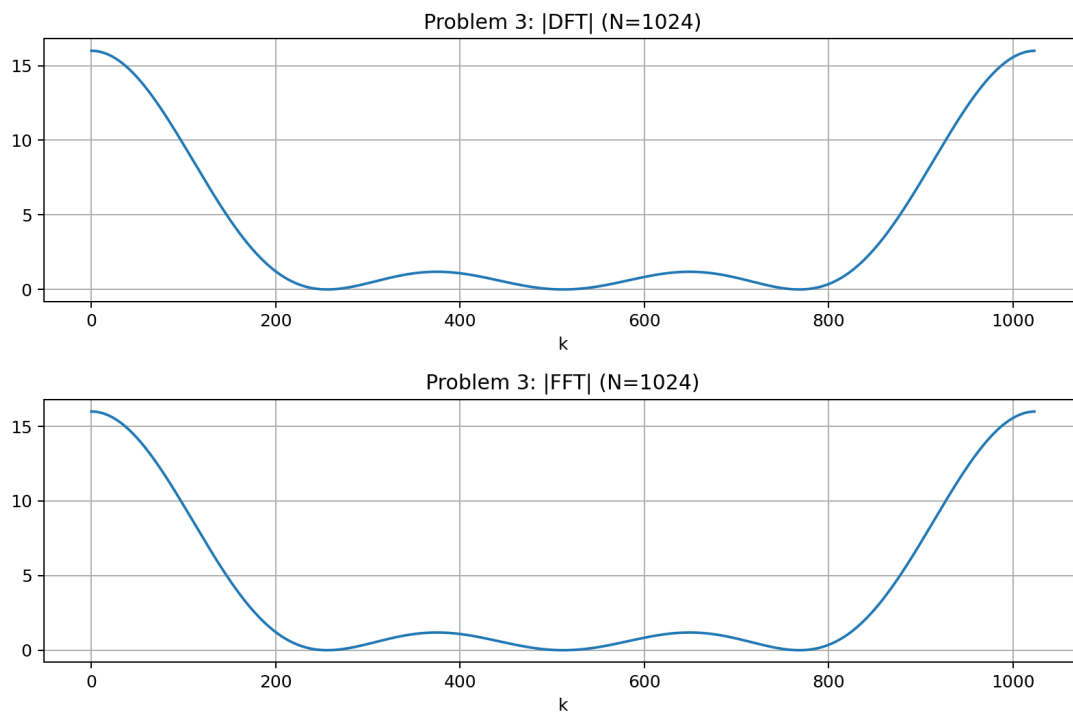


Figure 3: Magnitude spectra from direct DFT and FFT ($N = 1024$)

4 Problem 4: Effect of Zero-Padding on Spectral Resolution

4.1 Task

Append 8 zeros to get a 16-point sequence and compare spectra:

$$X_8[k] = \text{FFT}\{x[n], 8\}, \quad X_{16}[k] = \text{FFT}\{x[n], 0, \dots, 0, 16\}$$

4.2 Observation

Zero-padding does not add new signal information, but increases frequency-domain sampling density. The envelope is the same, while spectral curve appears more detailed.

4.3 Figure

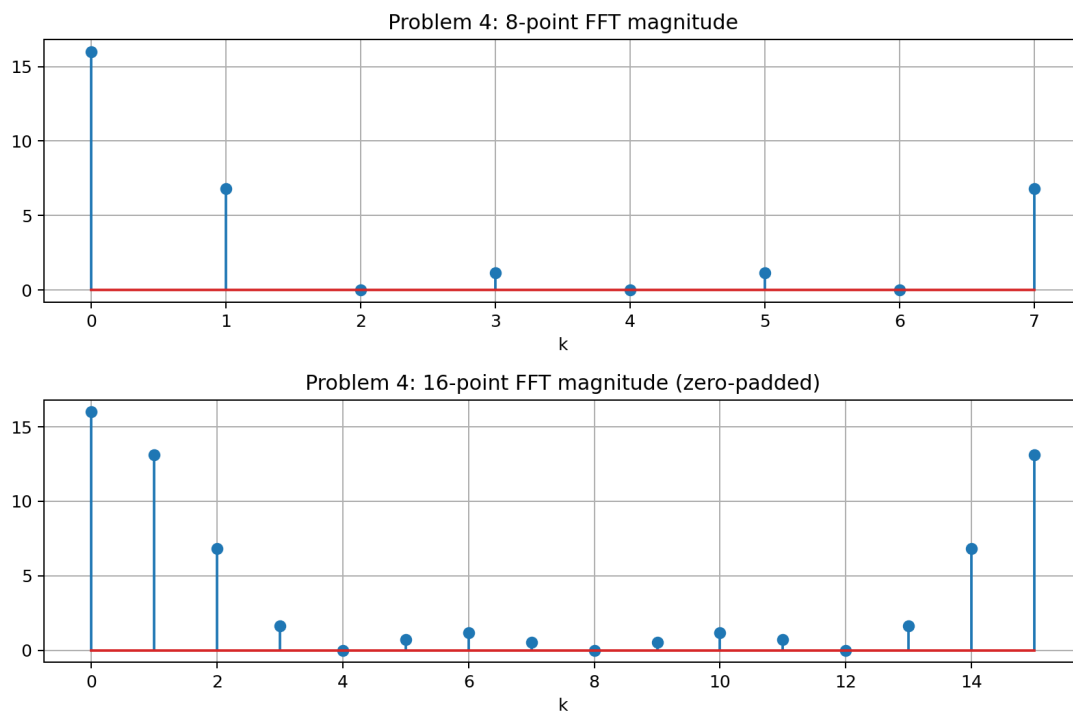


Figure 4: 8-point vs 16-point FFT magnitude (after zero-padding)

5 Problem 5: Low-Pass Filtering for Speech Signal Reconstruction

5.1 Given

$$x_{\text{clean}}[n] = \sin(0.2\pi n) + 0.5 \sin(0.7\pi n)$$

$$x[n] = x_{\text{clean}}[n] + 0.2\varepsilon[n], \quad \varepsilon[n] \sim \mathcal{N}(0, 1)$$

5.2 Method

Compute $X[k] = \text{FFT}\{x[n]\}$, keep only the lowest 10% frequency bins (and their conjugate-symmetric bins), set others to zero:

$$X_{\text{lp}}[k] = \begin{cases} X[k], & \text{low-frequency bins} \\ 0, & \text{otherwise} \end{cases}$$

Then reconstruct:

$$y[n] = \Re\{\text{IFFT}(X_{\text{lp}}[k])\}$$

5.3 Discussion

Low-pass filtering suppresses high-frequency noise and smooths the waveform. However, it may also attenuate useful high-frequency signal content, so denoising and detail preservation must be balanced.

5.4 Figures

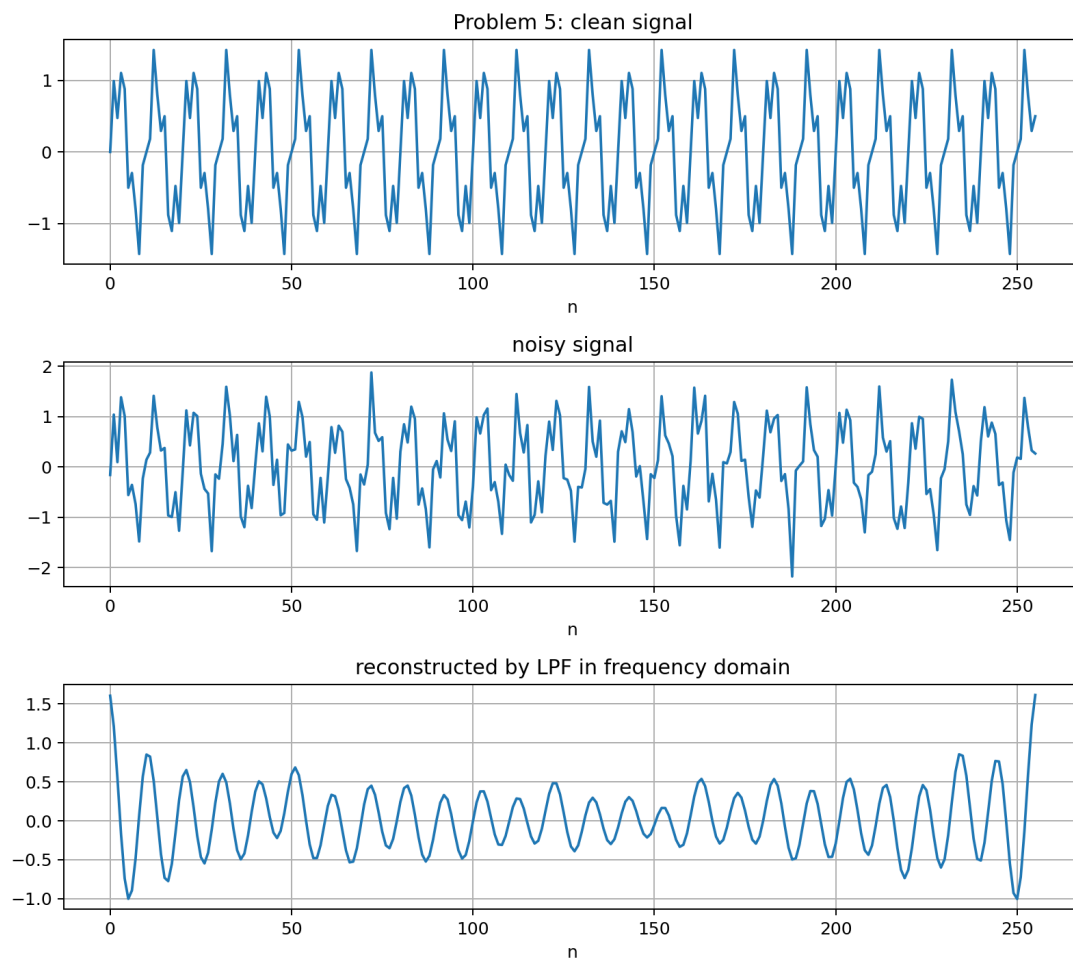


Figure 5: Clean signal, noisy signal, and reconstructed signal

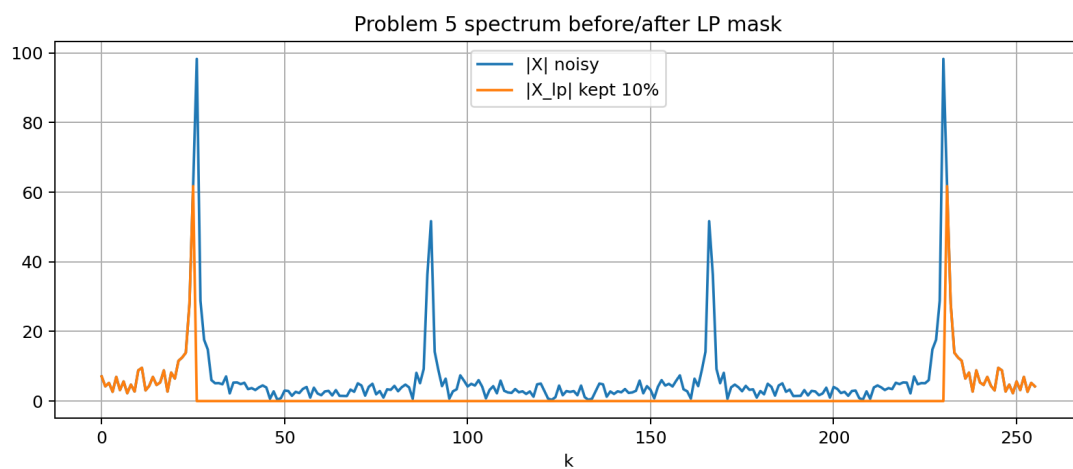


Figure 6: Spectrum before and after low-pass masking

6 Conclusion

This lab verifies three practical DSP points:

1. DFT provides frequency-domain representation of finite signals.
2. FFT is significantly faster than direct DFT for large transform sizes.
3. Frequency-domain masking enables denoising and reconstruction, but introduces a trade-off between noise suppression and signal detail.